

Algorithms Problem Solving

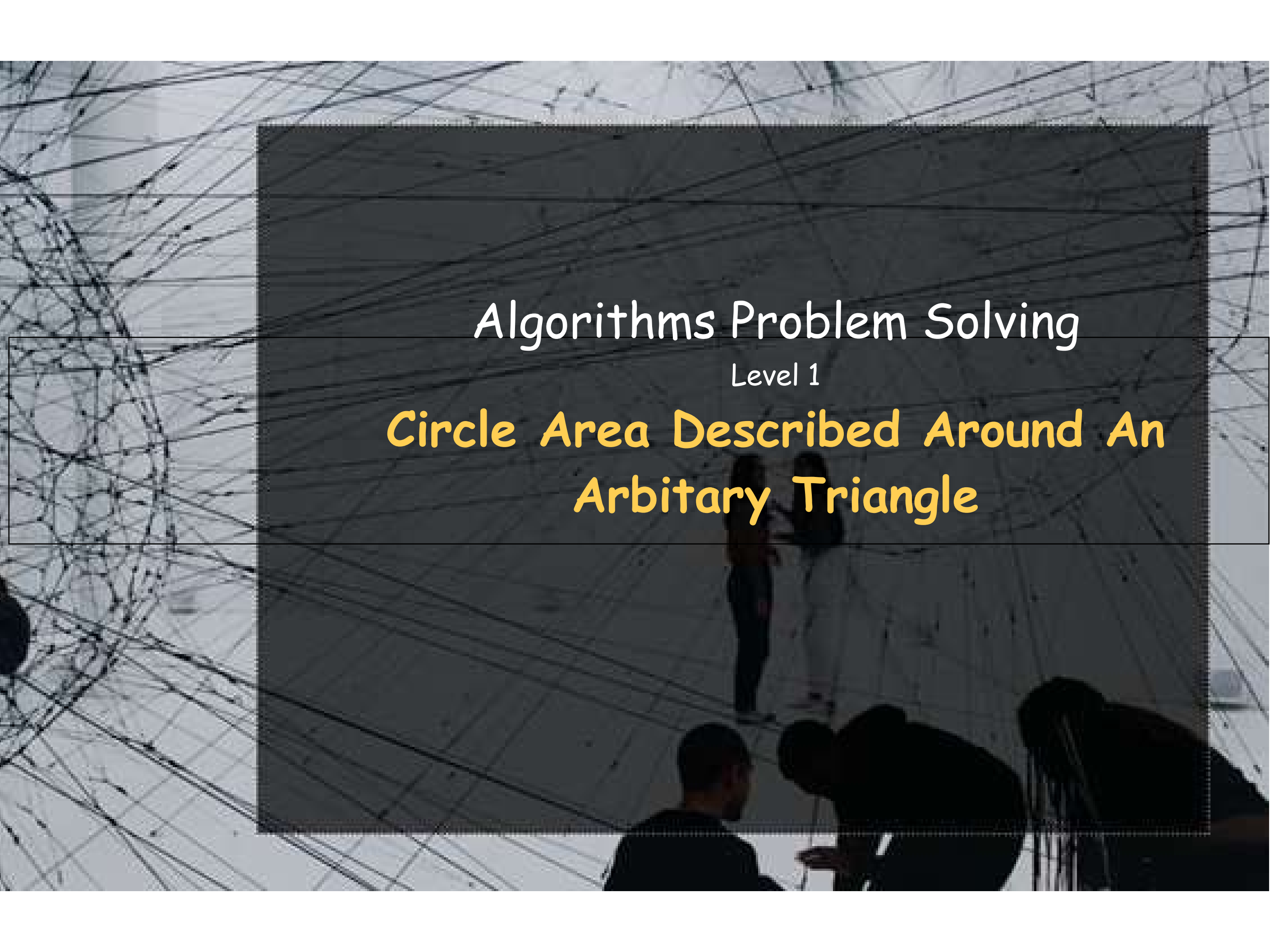
Level 1

Circle Area Described
Around An Arbitrary
Triangle

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Circle Area Described Around An Arbitrary Triangle



Definition

- ◆ This program calculates the area of a circle circumscribed around an arbitrary triangle.
- ◆ The triangle is defined by its three side lengths: A , B , and C .
- ◆ The program first calculates the triangle's circumradius, then uses it to calculate the circle area.

Explanation

- ◆ ``readTriangleData()`` reads the three triangle sides and stores them in *A*, *B*, and *C* using reference parameters.
- ◆ ``P = (A + B + C) / 2`` calculates the semi-perimeter of the triangle.
- ◆ ``T`` represents the circumradius of the triangle.
- ◆ The circumradius is calculated using the three side lengths and the semi-perimeter.
- ◆ ``M_PI * pow(T, 2)`` calculates the area of the circle using the radius *T*.
- ◆ ``printResult()`` displays the calculated circle area.
- ◆ ``printResult(CircleAreaByArbitraryTriangle(A, B, C))`` passes the returned circle area directly to the print function.

Steps in main()

- 1 Read the three triangle sides A , B , and C .
- 2 Calculate the semi-perimeter P .
- 3 Calculate the circumradius T .
- 4 Calculate the circle area using $\pi \times T^2$.
- 5 Return the calculated area.
- 6 Print the result.

💡 Example

◆ If the user enters:

$A = 5$

$B = 6$

$C = 7$

◆ The program calculates the circumradius and then the area of the circumscribed circle.

➡ Result

Circle Area = 40.088

Note

- ◆ The circle described around a triangle is called the circumscribed circle, or circumcircle.
- ◆ The triangle's three vertices lie on the circumference of this circle.
- ◆ ``sqrt()`` calculates the square root, while ``pow(T, 2)`` calculates T^2 .

⊘ Warning

- ◆ The three side lengths must form a valid triangle.
- ◆ The code does not validate the triangle inequality before performing the calculation.



Comparison

Circumscribed Circle

- ✓ The circle surrounds the triangle.
- ✓ All three triangle vertices lie on the circle.
- ✓ The triangle sides are used to calculate the circle's radius.

Inscribed Circle

- ✓ The circle is inside the triangle.
- ✓ The circle touches the three sides of the triangle.
- ✓ Its radius is calculated differently.

✦✦ Summary

- ✓ The program calculates the area of a circumcircle around an arbitrary triangle.
- ✓ The three triangle sides A , B , and C are used as input.
- ✓ The semi-perimeter is calculated first.
- ✓ The circumradius is then used to calculate the circle area.
- ✓ The three sides must form a valid triangle.

Thank you 😊